QML Lab • research project 02

Trainable Quantum Kernels

Learning a feature map whose kernel matches the labels, and measuring what that buys.

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Quadrant	CQ (classical data, quantum processor)	**Track**	Kernels
Difficulty	Medium	**Tools**	PennyLane, scikit-learn
Builds on	Modules 3 and 6 of the QML Lab	**Workspace**	`qml_projects/trainable-quantum-kernels/`

Summary. A fixed feature map is a guess about which data points should look similar. This project makes the feature map trainable and optimises it to maximise kernel–target alignment, a cheap proxy for how well an SVM will perform. The question is whether alignment training improves test accuracy over a fixed ZZ map and a tuned RBF kernel, and at what cost in circuit evaluations and overfitting.

1 Background and key concepts

1.1 Quantum embedding kernels

A quantum embedding kernel uses a parameterised feature map $|\psi_\theta(x)\rangle = U_\theta(x)\,|0\rangle$,

$$\kappa_\theta(x, x') = |\,\langle\psi_\theta(x)|\psi_\theta(x')\rangle\,|^2. \tag{1}$$

With fixed θ this is an ordinary quantum kernel (Module 6). Training θ adapts the geometry of the feature space to the task. Hubregtsen et al. (2022) studied this on near-term hardware, including the effect of noise on the kernel matrix.

1.2 Kernel–target alignment

Training a full SVM inside an optimisation loop is expensive. Kernel–target alignment (Cristianini et al., 2001) measures how close the Gram matrix K is to the ideal kernel $yy^\top$ for labels $y \in \{-1, +1\}^N$:

$$A(K, y) = \frac{\langle K, yy^\top\rangle_F}{\|K\|_F\,\|yy^\top\|_F} = \frac{\sum_{ij} y_i y_j K_{ij}}{N\sqrt{\sum_{ij} K_{ij}^2}}. \tag{2}$$

$A = 1$ means same-class points are maximally similar and different-class points dissimilar. Gradients $\partial A/\partial\theta$ follow from parameter-shift gradients of each kernel entry (Module 3). Each step uses a random batch of B points, so it costs $O(B^2)$ kernel evaluations rather than $O(N^2)$.

1.3 Inductive bias

Kübler et al. (2021) show that quantum kernels generalise only if their inductive bias, the functions they find easy to represent, matches the task; very expressive embeddings spread

their capacity over an exponentially large space and fail to generalise. Training the embedding is one way to create a useful bias, but it also adds parameters that can overfit.

2 Research questions

The project is open-ended: these questions are the starting point, not the finish line. Add new ones to the README and the journal as they come up.

RQ1 Does higher training alignment translate into higher test accuracy, and where does the correlation break down?

RQ2 How many trainable parameters can the embedding have before the trained kernel overfits?

RQ3 Can alignment be trained on a small subset (10–20 %) of the data and still transfer to the full training set?

3 Method

3.1 Data

Two moons and concentric circles (scikit-learn) for intuition, then breast cancer (30 features, PCA to 4) and a two-class subset of Iris. Features are standardised and scaled to $[0, \pi]$. Every data set gets a fixed train/validation/test split per seed.

3.2 Models

A layered embedding on $n = 4$ qubits: each layer applies angle encoding of x (features repeated over qubits if $d < n$), trainable R_Z/R_Y rotations and a ring of CNOTs. θ is trained by gradient ascent on the alignment of random batches ($B = 8$–16); the final kernel matrix is then handed to a precomputed-kernel SVM.

3.3 Baselines

The same embedding with random, untrained θ; the fixed ZZ feature map; and an RBF kernel whose width is also chosen by alignment, so that both sides get the same tuning method.

3.4 Metrics

- Training and test alignment; test accuracy of the SVM.
- Number of kernel evaluations used for training.
- Alignment and accuracy as a function of training steps (learning curves).

4 Planned experiments

Each experiment becomes one `[[experiments]]` entry in the workspace, so its runs, parameters and metrics are stored and compared in the portal.

ID	Experiment	What varies	Main metrics
E1	Alignment training on two moons	training steps (0–300)	alignment, test accuracy
E2	Embedding size	layers $L \in \{1, 2, 3, 4\}$	train/test gap
E3	Batch size	$B \in \{4, 8, 16, 32\}$	alignment noise, cost
E4	Real data	breast cancer, Iris	test accuracy vs RBF
E5	Subset transfer	alignment trained on 10% of data	test accuracy

Table 1: Experiment plan. Results go into the journal and the results log (section 9).

5 Running it in the QML Lab

The workspace lives in `qml_projects/trainable-quantum-kernels/`. Scripts import three helpers from `qml_lab/plots.py`:

- `params()` reads the parameters from the portal's run form (defaults when run by hand);
- `save()` stores figures with the run;
- `log_metrics()` records numbers that appear in the run table.

```
qml_projects/trainable-quantum-kernels/
  project.toml          metadata + experiments
  README.md             goal, open questions, ideas
  experiments/          runnable scripts
  docs/main.tex         this document
```

First experiment: This experiment is *planned*: write the script, add the block below to `project.toml`, then run `python manage.py sync_projects`.

```
[[experiments]]
slug = "alignment-moons"
title = "Alignment-trained embedding kernel on two moons"
script = "experiments/e01_alignment_moons.py"
figure = "alignment.png"

[experiments.params]
n_qubits = 4
n_layers = 2
steps = 200
batch = 10
lr = 0.05
seed = 0
```

```
# from the portal folder: pick up changes to project.toml
python manage.py sync_projects
cd qml_projects/trainable-quantum-kernels
# run by hand with the default parameters
PYTHONPATH=../../qml_lab python experiments/e01_alignment_moons.py
```

6 Quality bar

- A classical baseline with a similar number of trainable parameters, tuned with the same care (Bowles et al., 2024).
- At least five random seeds per configuration, reported as mean $\pm$ standard deviation.
- How results change with the number of qubits.
- At least one run under a realistic noise model.
- Choose hyperparameters on a validation split; never tune alignment on the test set.

7 Open directions

- Centred alignment, which is less sensitive to class imbalance.
- Multi-class alignment with one-hot label kernels.
- Effect of hardware noise on trained kernels, following Hubregtsen et al. (2022).
- Compare alignment with the geometric difference of Huang et al. (2021) as a selection criterion.

8 Risks and limitations

- Alignment is a proxy: high alignment on training batches can coexist with poor test accuracy.
- Each step costs $O(B^2)$ kernel evaluations, each needing parameter-shift gradients.
- Too many embedding parameters overfit small data sets.

9 Results log

Summarise findings here as they become stable; day-to-day notes belong in the portal journal.

Date	Run	Finding	Next step

References

Bowles, J., Ahmed, S. and Schuld, M. (2024) 'Better than classical? The subtle art of benchmarking quantum machine learning models', Preprint, arXiv:2403.07059. Available at: https://arxiv.org/abs/2403.07059.

Cristianini, N., Shawe-Taylor, J., Elisseeff, A. and Kandola, J. (2001) 'On kernel-target alignment', in *Advances in Neural Information Processing Systems 14 (NIPS 2001)*, pp. 367–373.

Huang, H.-Y., Broughton, M., Mohseni, M., Babbush, R., Boixo, S., Neven, H. and McClean, J. R. (2021) 'Power of data in quantum machine learning', *Nature Communications*, 12, 2631. Available at: https://doi.org/10.1038/s41467-021-22539-9.

Hubregtsen, T., Wierichs, D., Gil-Fuster, E., Derks, P.-J. H. S., Faehrmann, P. K. and Meyer, J. J. (2022) 'Training quantum embedding kernels on near-term quantum computers', *Physical Review A*, 106(4), 042431. Available at: https://doi.org/10.1103/PhysRevA.106.042431.

Kübler, J. M., Buchholz, S. and Schölkopf, B. (2021) 'The inductive bias of quantum kernels', in *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, pp. 12661–12673.